Office Placement Application (OPA) README

📌 Project Overview

The Office Placement Application (OPA) is a web-based platform designed to digitally track and manage faculty office assignments across university buildings. The system replaces manual spreadsheets and fragmented records with a centralized, searchable, and auditable office management solution.

🎯 Objective

To digitally manage and track faculty office assignments while ensuring:
- Centralized and up-to-date office records
- Clear occupant and capacity tracking
- Conflict detection and resolution
- Exportable reports for planning purposes
- Historical record of office occupancy

🚀 Key Features

1. Office Database
- Store office details: office number, building, floor, and capacity.
- Maintain a record of current occupants (faculty/staff).

2. Occupant Management
- Assign and unassign faculty/staff to offices.
- Track occupancy history (previous occupants per office).

3. Search & Filtering
- Search by office number, occupant name, or department.
- Filter by building, floor, or availability status.

4. Conflict Detection
- Flag offices that exceed capacity.
- Alert admins when assignment conflicts are detected.

5. Admin Panel
- Update office assignments and details.
- Manage buildings, floors, and room records.
- View and resolve flagged conflicts.

6. Exportable Reports
-Generate reports filtered by department, building, or occupancy status.
- Export in PDF or Excel format for planning purposes.

7. Office History Tracking
- Log all assignment changes with timestamps.
- View full occupancy history per office.

🏗️ System Architecture

Frontend
- React.js with advanced search and filtering
- Role-based dashboards (Admin / Viewer)
- Optional: Campus map integration with office visualization

Backend
- Django REST API

Database
- PostgreSQL with relational mapping (offices, occupants, buildings, history)

Authentication
- University Single Sign-On (SSO)

👥 User Roles
- Faculty / Staff (view own assignment)
- Department Admin (manage department offices)
- System Administrator (full access)

🔐 Security
- SSO authentication
- Role-based access control
- Audit logs for all assignment changes

📦 Deployment
- Cloud-based deployment
- Scalable infrastructure
- PostgreSQL with relational data integrity

⚡ Getting Started

Note: Detailed setup instructions will be added as the project structure is finalized. The steps below reflect the expected development workflow.

Prerequisites
- Node.js >= 18
- Python >= 3.11
- PostgreSQL >= 14

Steps:

1. Clone the repository

```
git clone https://github.com/your-org/office-placement-app.git
  cd office-placement-app
```

2. Set up the backend

```
  cd backend
  pip install -r requirements.txt
  python manage.py migrate
  python manage.py runserver
```

3. Set up the frontend

```
  cd frontend
  npm install
  npm start
```

📈 **Future Enhancements (Optional)**

- Interactive campus map with clickable office visualization
- Automated assignment suggestions based on department and availability
- Integration with HR system for staff data sync
- Mobile-friendly interface

📝 **Note:**

This README reflects the current state of the project at its initial planning phase.
It will be continuously updated as development progresses — including setup details, API references, environment configuration, and contribution guidelines.